

Instructions:

- Make sure to show each step of your solution for full credit - correct answers without work will not receive credit. Some questions may remind you to show your work, but you should assume each question requires you to justify your steps by making your thought process clear in your solution. You can use the solutions we write in class as an indication of how much work you are expected to show. You can ask the instructor if it is not clear how much work you are expected to show on a particular problem.
- Write all of your work and answers on this document. If you need extra paper, please ask the instructor.
- Indicate your affirmation of the following statement by signing below.

I did not give or receive any unauthorized assistance on this exam. I understand that the penalty for academic dishonesty is an F for the course and possible suspension or expulsion from the university.

Signature: _____

Derivatives

$$\begin{aligned}\frac{d}{dx}(\sin^{-1} x) &= \frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\cos^{-1} x) &= -\frac{1}{\sqrt{1-x^2}} \\ \frac{d}{dx}(\tan^{-1} x) &= \frac{1}{1+x^2} \\ \frac{d}{dx}(\sinh^{-1} x) &= \frac{1}{\sqrt{1+x^2}} \\ \frac{d}{dx}(\cosh^{-1} x) &= \frac{1}{\sqrt{x^2-1}} \\ \frac{d}{dx}(\tanh^{-1} x) &= \frac{1}{1-x^2}\end{aligned}$$

Identities

$$\begin{aligned}\cos^2 x + \sin^2 x &= 1 \\ \tan^2 x + 1 &= \sec^2 x \\ \sin^2 x &= \frac{1}{2}(1 - \cos(2x)) \\ \cos^2 x &= \frac{1}{2}(1 + \cos(2x)) \\ \cosh^2 x - \sinh^2 x &= 1 \\ 1 - \tanh^2 x &= \operatorname{sech}^2 x\end{aligned}$$

Integrals

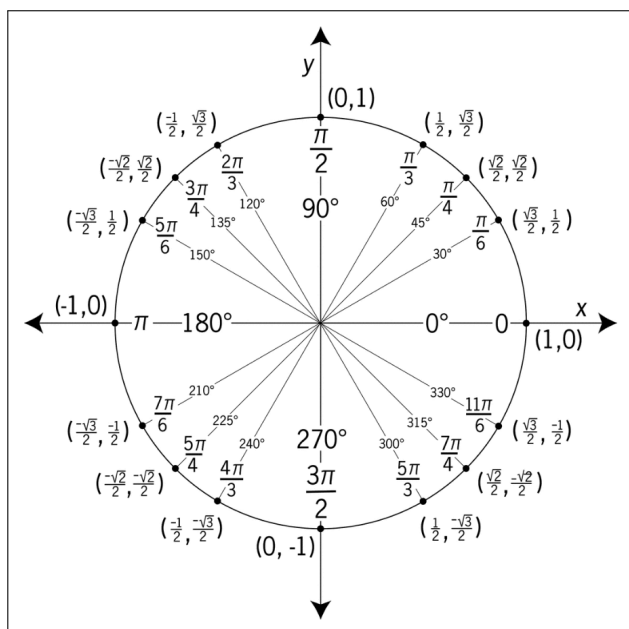
$$\begin{aligned}\int u dv &= uv - \int v du \\ \int \tan x dx &= \ln |\sec x| + C \\ \int \sec x dx &= \ln |\sec x + \tan x| + C \\ \int \frac{dx}{x^2 + a^2} &= \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C\end{aligned}$$

Approx. integration

$$\begin{aligned}|E_T| &\leq \frac{K(b-a)^3}{12n^2} \\ |E_M| &\leq \frac{K(b-a)^3}{24n^2} \\ |E_S| &\leq \frac{L(b-a)^5}{180n^4}\end{aligned}$$

Hyperbolic functions

$$\begin{aligned}\cosh(x) &= \frac{e^x + e^{-x}}{2} \\ \sinh(x) &= \frac{e^x - e^{-x}}{2} \\ \sinh^{-1} x &= \ln(x + \sqrt{x^2 + 1}) \\ \cosh^{-1} x &= \ln(x + \sqrt{x^2 - 1}) \\ \tanh^{-1} x &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)\end{aligned}$$

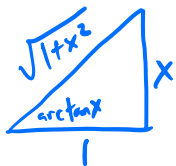


A1 - Inverse trigonometric and hyperbolic functions (inspired by Brightspace problem 9 and past final worksheet problem 1)

(a) Find an equivalent expression for

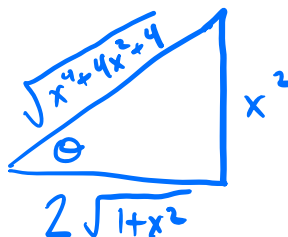
$$\sin \left(\underbrace{\operatorname{arccot} \left(\frac{2 \sec(\arctan x)}{x^2} \right)}_{\theta} \right)$$

that involves no trigonometric or inverse trigonometric functions.



$$\sec(\arctan x) = \sqrt{1+x^2}$$

$$\frac{2 \sec(\arctan x)}{x^2} = \frac{2 \sqrt{1+x^2}}{x^2}$$



$$\sin(\theta) =$$

$$\boxed{\frac{x^2}{\sqrt{x^4+4x^2+4}}}$$

$$c^2 = (x^2)^2 + (2\sqrt{1+x^2})^2$$

$$c = \sqrt{x^4 + 4x^2 + 4}$$

(b) Differentiate $y = \arctan(\sinh(2x))$ and simplify. Your final answer should use only one hyperbolic function.

$$y' = \frac{1}{1 + \sinh^2(2x)} \cdot \cosh(2x) \cdot 2 = \frac{1}{\cosh^2(2x)} \cdot \cosh(2x) \cdot 2$$

$$= \boxed{2 \operatorname{sech}(2x)}$$

(c) Evaluate $\lim_{x \rightarrow -\infty} \frac{e^{-x}}{\sinh x}$

$$\lim_{x \rightarrow -\infty} \frac{e^{-x}}{\frac{e^x - e^{-x}}{2}} = \lim_{x \rightarrow -\infty} \frac{2e^{-x} \cdot \overset{e^x}{e^x}}{\overset{e^x}{e^x} - e^{-x}} = \lim_{x \rightarrow -\infty} \frac{2}{\cancel{e^{2x^0}} - 1} = \boxed{-2}$$

A2 - L'Hospital's rule (inspired by Brightspace problem 8 and past final worksheet problem 1)

Evaluate each of the following limits.

(a) $\lim_{x \rightarrow -\infty} (4e^{5x} + 7e^{2x})^{\frac{3}{x}}$

$$L = \lim_{x \rightarrow -\infty} (4e^{5x} + 7e^{2x})^{\frac{3}{x}}$$

$$\ln(L) = \lim_{x \rightarrow -\infty} \frac{3}{x} \cdot \ln(4e^{5x} + 7e^{2x}) = \lim_{x \rightarrow -\infty} \frac{3 \ln(4e^{5x} + 7e^{2x})}{x}$$

$$\stackrel{H}{=} \lim_{x \rightarrow -\infty} \frac{3 \cdot \frac{20e^{5x} + 14e^{2x}}{4e^{5x} + 7e^{2x}} \cdot \frac{e^{2x}}{e^{2x}}}{1} = \lim_{x \rightarrow -\infty} 3 \cdot \frac{20e^{3x} + 14}{4e^{3x} + 7} = 3 \cdot \frac{14}{7} = 6$$

$$\rightarrow \ln(L) = 6 \rightarrow L = \boxed{e^6}$$

(b) $\lim_{x \rightarrow 0} \frac{\ln(1+x) - \sin x - \cos x + 1 - \frac{1}{2}x^3}{x^4}$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{1+x} - \cos x + \sin x - \frac{3}{2}x^2}{4x^3}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\frac{1}{(1+x)^2} + \sin x + \cos x - 3x}{12x^2}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\frac{2}{(1+x)^3} + \cos x - \sin x - 3}{24x}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\frac{6}{(1+x)^4} - \sin x - \cos x}{24} = \boxed{\frac{-7}{24}}$$

A3 - Integration techniques I (inspired by part 2 of the class notes and past final worksheet problem 4)

Evaluate the following integrals. You do not need to simplify your answers.

(a) $\int \left(\frac{\ln x}{x} \right)^2 dx$

$$u = (\ln x)^2 \quad v = -\frac{1}{x}$$
$$du = 2 \cdot \ln x \cdot \frac{1}{x} dx \quad dv = \frac{1}{x^2} dx$$

$$= \frac{-(\ln x)^2}{x} - \int \frac{1}{x} \cdot 2 \cdot \ln x \cdot \frac{1}{x} dx = \frac{-(\ln x)^2}{x} + 2 \int \frac{\ln x}{x^2} dx$$

$$u = \ln x \quad v = -\frac{1}{x}$$
$$du = \frac{1}{x} dx \quad dv = \frac{1}{x^2} dx$$

$$= \frac{-(\ln x)^2}{x} + 2 \left(\ln x \cdot -\frac{1}{x} - \int \frac{1}{x} \cdot \frac{1}{x} dx \right)$$

$$= \frac{-(\ln x)^2}{x} - \frac{2 \ln x}{x} + 2 \int \frac{1}{x^2} dx = \boxed{\frac{-(\ln x)^2}{x} - \frac{2 \ln x}{x} - \frac{2}{x} + C}$$

(b) $\int \frac{x^2}{(4-x^2)^{\frac{3}{2}}} dx$

$$x = 2 \sin \theta$$
$$dx = 2 \cos \theta d\theta$$



$$\int \frac{(2 \sin \theta)^2}{(4 - 4 \sin^2 \theta)^{\frac{3}{2}}} \cdot 2 \cos \theta d\theta = \int \frac{4 \sin^2 \theta}{(4 \cos^2 \theta)^{\frac{3}{2}}} \cdot 2 \cos \theta d\theta = \int \frac{\sin^2 \theta}{\cos^3 \theta} d\theta = \int \tan^2 \theta d\theta$$

$$= \int (\sec^2 \theta - 1) d\theta$$

$$= \tan \theta - \theta + C$$

$$= \boxed{\frac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C}$$

A4 - Integration techniques II (inspired by part 3 of the class notes and Brightspace problem 5)

(a) Evaluate $\int \frac{10x+40}{(x-1)(x^2+4)^2} dx$. You do not need to simplify your answer.

$$= \frac{A}{x-1} + \frac{Bx+C}{x^2+4} + \frac{Dx+E}{(x^2+4)^2}$$

$$10x+40 = A(x^2+4)^2 + (Bx+C)(x-1)(x^2+4) + (Dx+E)(x-1)$$

$$A(x^4+8x^2+16) + (Bx+C)(x^3-x^2+4x-4) + (Dx+E)(x-1)$$

$$x=1: 50 = A(1^2+4)^2 \rightarrow A=2$$

$$x^4: 0 = A+B \rightarrow B=-2$$

$$x^3: 0 = -B+C \rightarrow C=-2$$

$$x^2: 0 = 8A+4B-C+D \rightarrow D=10$$

$$x: 10 = -4B+4C-D+E \rightarrow E=0$$

$$\int \left(\frac{2}{x-1} + \frac{-2x-2}{x^2+4} + \frac{-10x}{(x^2+4)^2} \right) dx$$

$$2 \ln|x-1| - \ln|x^2+4| - \frac{2}{2} \tan^{-1}\left(\frac{x}{2}\right) + \frac{5}{x^2+4} + C$$

(b) Determine if the integral $\int_0^\infty \frac{\sin(x)+2}{x^2+\sqrt{x}} dx$ converges or diverges.

$$\int_0^\infty \frac{\sin(x)+2}{x^2+\sqrt{x}} dx = \int_0^1 \frac{\sin(x)+2}{x^2+\sqrt{x}} dx + \int_1^\infty \frac{\sin(x)+2}{x^2+\sqrt{x}} dx$$

$$\int_0^1 \frac{\sin(x)+2}{x^2+\sqrt{x}} dx \leq \int_0^1 \frac{3}{\sqrt{x}} dx = \lim_{a \rightarrow 0^+} 6\sqrt{x} \Big|_0^1 = 6$$

$$\int_1^\infty \frac{\sin(x)+2}{x^2+\sqrt{x}} dx \leq \int_1^\infty \frac{3}{x^2} dx = \lim_{b \rightarrow \infty} -\frac{3}{x} \Big|_1^b = 3$$

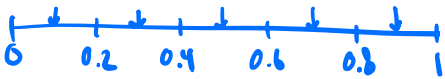
$$\text{So } \int_0^\infty \frac{\sin(x)+2}{x^2+\sqrt{x}} dx \leq 9 \rightarrow \text{converges}$$

A5 - Approximate integration (inspired by the class notes and Brightspace problem 7)

Consider the integral $I = \int_0^1 e^{x^3} dx$. For each of the questions below, you do not need to simplify your answers.

- (a) Use the Midpoint Rule with $n = 5$ to approximate I .

$\Delta x = 0.2$



$$0.2(e^{0.1^3} + e^{0.3^3} + e^{0.5^3} + e^{0.7^3} + e^{0.9^3})$$

- (b) How large should n be so that the Midpoint Rule approximates I to within 0.001?

$$|E_n| \leq \frac{k \cdot (1-0)^3}{24n^2}$$

$$f(x) = e^{x^3}$$

$$f'(x) = 3x^2 e^{x^3}$$

$$f''(x) = 6x e^{x^3} + 9x^4 e^{x^3} \leq 15e = k \text{ at } x=1$$

$$\frac{15e}{24n^2} \leq 0.001$$

$$\rightarrow n \geq \sqrt{\frac{15e}{24 \cdot 0.001}}$$

- (c) Use Simpson's Rule with $n = 6$ to approximate I .

$\Delta x = \frac{1}{6}$

$$\frac{1}{3} \left(e^{0^3} + 4e^{\left(\frac{1}{6}\right)^3} + 2e^{\left(\frac{2}{6}\right)^3} + 4e^{\left(\frac{3}{6}\right)^3} + 2e^{\left(\frac{4}{6}\right)^3} + 4e^{\left(\frac{5}{6}\right)^3} + e^{\left(\frac{6}{6}\right)^3} \right)$$

- (d) The error from the approximation in part (c) is about 0.0008. What would we expect the error to be if we used Simpson's Rule with $n = 12$?

When n doubles, $|E|$ decreases by a factor of $\frac{1}{2^4} = \frac{1}{16}$,
 so the error would be about $\frac{0.0008}{16}$